CETRARO LECTURE 2

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Recap: we were considering an arrangement of hyperplanes $\mathcal{H}$, with a chosen base chamber B. We ordered the chambers by inclusion of $s_B(\cdot)$, the hyperplanes separating the chamber from B. The cover relations were crossing from a chamber to an adjacent one. The resulting order $L_{\mathcal{H}}$ is a lattice if all the regions are simplicial (though this is more than we actually need).

One example was the reflection arrangement, in which case we can identify the chambers with the elements of the corresponding finite Coxeter group, and the order with weak order.

We defined a way to split hyperplanes. When multiple hyperplanes intersect, we say that those two are basic which are adjacent to the region containing the base chamber.

We define a way to cut a hyperplane into shards. From H, we remove $H \cap H'$ whenever at that intersection H is not basic.

We can also think in terms of normal vectors: H_γ, with normal vector γ, has a codimension 2 intersection with H_α, H_β provided α, β, γ are linearly dependant. H_γ is cut if γ is a positive combination of α and β. (Signs chosen uniformly – pointing **towards** B. (This is opposite to what I said yesterday, I think, but it's what's consistent with the conventions I want.)

1.4. **Shards are in bijection with join-irreducibles.** Let S be a shard of some hyperplane H, and consider the chambers within S. We put an order on them by considering the corresponding $\mathcal{H}$-chambers. The goal will be to show that a minimal chamber in S corresponds to a join-irreducible $\mathcal{H}$-chamber, and that there is exactly one minimal element with respect to this order, thus giving us what we want.

The approach I am taking here is closer to that of [M] than the approach of Reading. Reading's approach uses some more powerful lattice theory, which is lovely, but I would either have had to not prove a bunch of things, or else develop a lot of machinery.

Lemma 3. *If a chamber x of S is minimal, then the corresponding $\mathcal{H}$-chamber, X, is join-irreducible.*

Proof. Assume that the chamber X is not join-irreducible. Therefore, there are at least two ways to go down from it, crossing H and crossing some other hyperplane J. Draw the rank 2 picture for H and J. We can see that there is another chamber lower than X with H lying below it, and such that it lies in the same shard, because, H being basic in this arrangement, it is not cut. □

What are we really doing when we make this "rank 2" argument? We are saying: take a generic point in the codimension 2 intersection in question, and take a little two-dimensional slice transverse to it.

Lemma 4. *Adjacent chambers in S are comparable.*

Proof. We draw the rank 2 picture for the subspace where the chambers intersect. The hyperplane H must be basic, since otherwise the two chambers we are considering would belong to different shards. Now it is visible in the rank 2 picture. $\square$

The following lemma is slightly confusing to state, but useful.

Lemma 5. *The order defined on chambers of S by transitive closure of the induced order on adjacent pairs is the same as the induced order.*

The point of the lemma is that we could define another order on the chambers of S, just by taking adjacent pairs of chambers and ordering them according to the order on the corresponding $\mathcal{H}$-chambers, and then taking the transitive closure. This is somehow more in keeping with the idea of localizing our attention to S.

Proof. Suppose that we have x, y chambers in S, such that the corresponding $\mathcal{H}$-chambers are X, Y, with $X < Y$. There is therefore some hyperplane J defining a wall of X by which we can go up while staying below Y. This wall cuts S along a facet of x. Let z be the chamber in S on the opposite side of J from x, and Z the corresponding $\mathcal{H}$-chamber. Then

$$s(Z) = s(X) \cup \{\text{the hyperplanes cutting } H \text{ along } H \cap J\}.$$

But $s(Y)$ certainly contains $s(Z)$, so $Y > Z$, and we are done by induction. $\square$

Lemma 6. *If we have three chambers x, y, z in S with x covering y and z, then there is also a chamber $w \in S$ less than y, z.*

Proof. We draw the ... rank three picture! We see that X, the $\mathcal{H}$-chamber corresponding to x is at the very top (since there are three ways to go down from it). So the hyperplanes incident to it are all basic and don't ever get cut. The consequence is that H (restricted to this picture) is a full two-dimensional hyperplane. Its bottom region w, corresponds to the $\mathcal{H}$-chamber W, which is only separated from B by H. Thus, W is below Y and Z, so the same holds for w, y, z. $\square$

Lemma 7. *If a chamber m in S is minimal, then it is actually the minimum.*

Proof. Assume otherwise, so m is minimal but not the minimum. Since the poset of chambers of S is connected, there must be an element x which lies over m and also lies over some element z which does not lie over m. Choose x to be minimal among such elements, x must cover z. By Lemma 5, x also covers some y which lies over m. Now applying Lemma 6, we get a chamber w below z and y. Now w cannot lie over x, but y lies over w and over m. So our initial choice of x was not minimal. $\square$

Theorem 1. *The natural map from join-irreducibles to shards, sending a join-irreducible chamber J to the shard including the unique panel below J, is a bijection.*

Proof. For each shard S, we know that the order induced on its chambers has a unique minimum m (by Lemma 7). The corresponding $\mathcal{H}$-chamber, M is join-irreducible by Lemma 3. Any other chamber x in the shard is not minimal, so it has at least one lower neighbour in the poset of chambers in the shard, and therefore there is at least one way to descend from the corresponding $\mathcal{H}$-chamber X, other than crossing S. So X has at least two lower covers, and it is therefore not join-irreducible. $\square$

We will later need a lemma whose proof fits naturally here:

Lemma 8. *For a shard S, there is a unique minimal element among all the elements having an upper facet on S, and it is J_*, where J is the join-irreducible corresponding to S.*

Proof. Let U be a chamber with a lower facet on S. We have already seen that $J \leq U$, and in fact that there is a sequence of adjacent chambers $x_0, \ldots, x_r$ in H where, if we write X_i for the $\mathcal{H}$-chamber haivng x_i as its lower facet, then $X_i < X_{i+1}$. But looking at the corresponding rank two picture, we also see that if we write Y_i for the $\mathcal{H}$ chamber having x_i as its upper facet, the same analysis shows that $Y_i < Y_{i+1}$. The desired conclusion follows. $\square$

2. Shards and lattice quotients

An equivalence relation $\equiv_\Theta$ on a lattice L respects the lattice operations if $x \equiv_\Theta x'$ and $y \equiv_\Theta y'$ imply that $x \vee y \equiv_\Theta x' \vee y'$ and $x \wedge y \equiv_\Theta x' \wedge y'$. Obviously, this means that the lattice operations descend to well-defined operations $\vee$ and $\wedge$ on the equivalence classes.

Note that in a lattice L, we have $x \geq y$ iff $x \vee y = x$ iff $x \wedge y = y$. This makes clear how to define the lattice operations on $L/\equiv_\Theta$.

Proposition 2. *There is a lattice structure on the equivalence classes, where $[x]_\Theta \geq [y]_\Theta$ iff $[x]_\Theta \vee [y]_\Theta = [x]_\Theta$ iff $[x]_\Theta \wedge [y]_\Theta = [y]_\Theta$*

Proof. Exercise. $\square$

Lemma 9. *The equivalence classes of a lattice congruence are intervals.*

Proof. The join of all of the elements of an equivalence class must also lie in the same equivalence class, thus the equivalence class has a maximum element. The dual argument shows that each equivalence class has a minimum element. Further, by antisymmetry of the poset structure on the equivalence classes, any element in the interval must be in the equivalence class as well. $\square$

Given a lattice congruence $\equiv_\Theta$, we say that it *contracts* a cover relation $b \gtrdot a$ iff $b \equiv_\Theta a$. We say it contracts a join-irreducible j if and only if $j \equiv_\Theta j_*$.

Lemma 10. *The Hasse diagram of $L/\equiv_\Theta$ is obtained from that of L by contracting the cover relations contracted by $\equiv_\Theta$, and replacing multiple edges joining a given pair of (new) vertices by a single edge.*

Proof. Contracting the cover relations replaces equivalence classes (which are intervals by the previous lemma) by a single vertex, so the given procedure produces the right collection of vertices. Uncontracted cover relations certainly remain order relations. Suppose that a cover relation $X \gtrdot Y$ became a relation $[X] > [Y]$ which was not a cover relation, i.e., we have $[X] > [Z] > [Y]$ for some Z. But in this case, $Z \vee Y > Y$ and $X > Y$, so $(Z \vee Y) \wedge X \geq Y$. But $[(Z \vee Y) \wedge X] = Z$, so we have $X > (Z \vee Y) \wedge X > Y$, which is a contradiction. $\square$

So, let us analyze which cover relations from the lattice of chambers survive in the quotient.

Lemma 11. *If $X \gtrdot Y$ is separated by a shard that is contracted, then $X \gtrdot Y$ is contracted.*

Proof. Let J be the join-irreducible corresponding to the facet in question. We have $J \leq X$ and $J_* \leq Y$. Further, $J \vee Y = X$, but $J_* \vee Y = Y$. Since $J \equiv_\Theta J_*$, we must have $X \equiv_\Theta Y$. □

Lemma 12. *If $X \gtrdot Y$ is separated by a shard that is not contracted, then $X \gtrdot Y$ is not contracted.*

Proof. Let J be the join-irreducible corresponding to the separating shard. We prove the contrapositive. Suppose $X \equiv_\Theta Y$. Then $X \wedge J = J$, but $Y \wedge J = J_*$. So $J \equiv_\Theta J_*$. □

The previous discussion proves the following theorem.

Theorem 2. *The Hasse diagram for a lattice quotient of the lattice of $\mathcal{H}$-chambers is obtained by erasing the contracted shards, putting a vertex in each resulting region, and putting covering relations through all the remaining walls.*

2.1. **Which sets of shards can be contracted?** We have seen that a lattice quotient of $L_\mathcal{H}$ contracts some set of shards, which determine the quotient. It is natural to ask which sets of shards can be contracted.

We say that a shard S forces a shard T if any quotient in which S is contracted also has T contracted.

Looking at the rank two situation, we see that either of the shards corresponding to a basic hyperplane forces all the others, except the other basic hyperplane, while none of the other shards force any other shard.

By looking at the rank two configurations in $\mathcal{H}$, we obtain the following:

Lemma 13. *Near a generic point p in a codimension 2 intersection in $\mathcal{H}$, we have shards S_1 and S_2 containing a neighbourhood around p of lying of the basic hyperplanes H_1 and H_2 for this intersection. We have an additional two shards, with p on their boundary, for each of the other hyperplanes containing $H_1 \cap H_2$. Then each of S_1 and S_2 force all the additional shards.*

In the above situation, we say the shards S_1 and S_2 cut the additional shards.

This gives us a necessary condition for a set of shards to be contracted:

Proposition 3. *In order for a set of shards to be contracted by some lattice equivalence, it must be closed under the cutting relation defined above.*

Reading also proves the converse of this proposition, but we will not see the proof.

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